# Exp 9 : PERFORMANCE EVALUATION OF AM GENERATION AND DETECTION UNDER NOISY CHANNEL CONDITIONS USING MATLAB

**Objectives :**

i) To generate an Amplitude Modulated (AM) signal, introduce Additive White Gaussian Noise (AWGN), and analyze the effect of noise on the transmitted signal using MATLAB.

ii) To recover the original message signal from the noisy AM signal using envelope detection and compare the transmitted and recovered waveforms to evaluate the performance of the AM communication system.

**Code :**

```
clc;
clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;
fc = 5000;

% Message Signal
m = cos(2*pi*fm*t);
c = cos(2*pi*fc*t);

% AM Modulation
am = (1 + m).*cos(2*pi*fc*t);

% Add AWGN Noise
noisy = awgn(am,10,'measured'); % 10 dB SNR

% Envelope Detection
env = abs(hilbert(noisy));

% Display Results
figure;

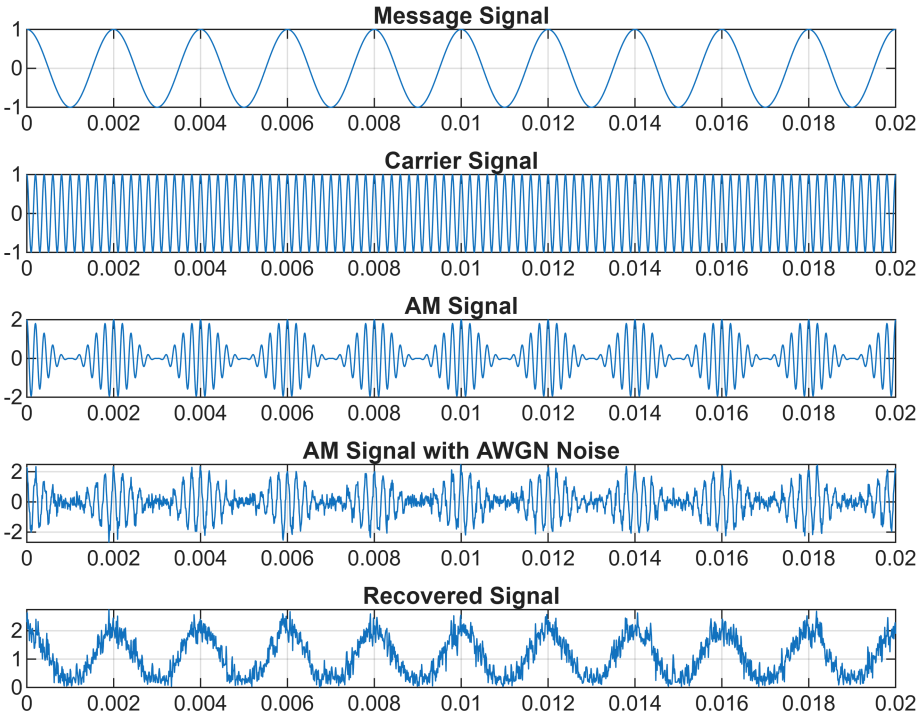
subplot(5,1,1)
plot(t,m)
title('Message Signal')
grid on
```

```
subplot(5,1,2)
plot(t,c)
title('Carrier Signal')
grid on

subplot(5,1,3)
plot(t,am)
title('AM Signal')
grid on

subplot(5,1,4)
plot(t,noisy)
title('AM Signal with AWGN Noise')
grid on

subplot(5,1,5)
plot(t,env)
title('Recovered Signal')
grid on
```




Result :

i) The AM signal was successfully generated in MATLAB, and the effect of Additive White

Gaussian Noise (AWGN) on the transmitted signal was observed through the noisy AM waveform.

ii) The original message signal was successfully recovered using envelope detection, demonstrating the effectiveness of the AM receiver in recovering the transmitted information in the presence of AWGN noise.